

AgroPal

Domain : AgriSense
(Team Ascend)

Understanding the Problem -

Smallholder farmers often lack immediate access to expert agricultural guidance and dependable market insights. As a result, crop issues may go undiagnosed and produce is frequently sold below its true market value. Existing agri-tech solutions are typically designed for large-scale farms and require constant internet connectivity, making them less practical for farmers who rely on basic smartphones and limited digital resources.

Identifying the Key Challenges -

- *Limited Access to Agronomy Experts* – Farmers often lack timely professional guidance for crop-related issues.
- *Lack of Real-Time Market Information* – Farmers struggle to determine the best time and place to sell their produce.
- *Late Detection of Crop Diseases* – Diseases and nutrient deficiencies are identified only after significant damage occurs.
- *Weather Uncertainty* – Unpredictable weather affects crop health, harvest planning, and profitability.
- *Low-Tech Farming Environment* – Many farmers rely on basic smartphones and limited internet connectivity, making advanced tools difficult to use.

Our Innovative Approach -

- *AI-Powered Crop Diagnosis* – Farmers can upload a crop photo and receive instant health analysis.
- *Easy-to-understand Guidance* – Diagnoses and recommendations are explained in simple, farmer-friendly language.
- *Integrated Weather Intelligence* – Provides localized weather insights to support farming decisions.
- *Smart Market Recommendations* – Suggests the best time and market to sell produce for maximum profit.
- *All-in-One Farm Companion* – Combines agronomy, weather forecasting, and market analysis in a single easy-to-use platform.

Architecture Overview -

- *Presentation Layer* – Streamlit-based user interface with Chat component.
- *Application Layer* – FastAPI backend managing requests and workflows.
- *AI Layer* – Disease classification model + Gemini-powered explanation engine.
- *Data Intelligence Layer* – Weather forecasting and market price analysis services.
- *Output Layer* – Personalized recommendations on crop health, farming actions, and optimal selling decisions.

Tech Stack Overview -

- Frontend - *Streamlit*
- Backend - *FastAPI*
- Programming language - *Python*
- AI Engine (for crop disease detection) - *CLIP*
- AI Engine (for suggestions and market analysis) - *Google Gemini*
- Chatbot - *Streamlit Chat UI*
- Image processing - *OpenCV*
- Weather Intelligence - *OpenWeather API*
- Development tool - *VS Code*

Why AgroPal?

- All-in-One Farm Companion.
- Photo-Based Crop Diagnosis.
- Weather-Aware Farming Advice.
- Market-Driven Selling Recommendations.
- Farmer-Friendly Conversational AI.

AgroPal – One AI companion, *EVERY* farming decision.

THANK YOU!